

Sentiment Analysis of Bengali Movie Reviews

Farhana Najnin Mahi

Department of CSE

20210104109

Dhaka, Bangladesh

farhana.cse.20210104109@aust.edu

Zarin Tasnim Roichi

Department of CSE

20210104111

Dhaka, Bangladesh

zarin.cse.20210104111@aust.edu

Afrain Akhter

Department of CSE

20210104113

Dhaka, Bangladesh

afrain.cse.20210104113@aust.edu

Maliha Akter Miti

Department of CSE

20210104122

Dhaka, Bangladesh

maliha.cse.20210104122@aust.edu

Abstract—Sentiment analysis (SA) is necessary for understanding user opinions and impacting decision-making in domains like review of products, feedback, and social media monitoring of social media. In this study, we have focused on Bangla sentiment classification using a dataset which contains 10,800 reviews, we have categorized them into positive(0) and negative(1) sentiments. We applied BanglaBERT, mBERT, XLM-RoBERTa (XLM-R), and Gemini, we also fine-tuned them by using LoRA (Low-Rank Adaptation) and PEFT (Parameter-Efficient Fine-Tuning) techniques. we have evaluated these models based on accuracy, precision, recall, and F1-score. Among these all model, BanglaBERT given the highest performance by achieving 97.90

Index Terms—Sentiment Analysis, Bengali NLP, BanglaBERT, mBERT, XLM-RoBERTa, LoRA, Text Classification, Deep Learning.

I. INTRODUCTION

Introduction: In today's internet-based communication environment, social media platforms and other e-commerce-based websites produce numerous amounts of text data. It is quite challenging to extract meaningful information from this data to understand users' behavior, preferences. Sentiment analysis, termed opinion mining, is a Natural Language Processing (NLP) task classified as positive, negative, or neutral. It has a great range of applications, and by taking feedback from customers they predict the market flow.

Though there are notable developments in sentiment analysis for widely used languages like English, there are not many studies on languages like Bangla due to the lack of availability of datasets, or maybe because of the lack of effective computational resources (Hoque et al., 2024)11†source. More than 230 million people speak in Bengali, it is world's 6th most spoken language, that makes sentiment analysis in Bengali crucial for businesses, researchers, and policymakers (Bhowmik et al., 2022)10†source. However, Sentiment analysis in the Bengali language leads to challenges like use of informal

language, dialect variation and the presence of code-mixed text (Tareq et al., 2023)11†source. Modern innovations in deep learning and transformer-based models have added a new dimension to sentiment analysis. Traditional approaches of machine learning approaches such as Support Vector Machines (SVM) and Naïve Bayes classifiers often face issues with complex language arrangements and rational understanding. With the creation of transformer based models like BERT, XLM-R, and multilingual BERT (mBERT), sentiment analysis models have achieved contemporary performance in text classification tasks (Sharma Nakrani, 2024)8†source. These models have reached a new height in achieving contextual dependencies that lead to great accuracy towards sentiment predictions, especially in low-resource languages like bangla. In this study, we have worked on sentiment predictions of movie reviews using a dataset that consists of 11,808 samples. The dataset includes user sentiments labeled as either positive (0) or negative (1), featuring a mixed range of opinions. We have implemented several transformer-based models, including BanglaBERT, mBERT, XLM-R, , Gemini to evaluate their effectiveness in classifying Bengali movie reviews. Our main goal of this research was to overcome the differences in Bengali sentiment analysis by Exploiting transformer models to improve classification accuracy and provide a benchmark for future studies in this domain.

The rest of the paper structure as follows: Section II represents a review of related works on Bengali sentiment analysis. Section III tells about the dataset and preprocessing techniques that we have used. Section IV discussed the experimental set up. Section V comparatively analyses the results obtained from different models. Finally, Section VI gives a conclusion of the paper including key findings and future research options.

II. RELATED WORK

Sentiment analysis in Bengali has really picked up pace recently, mainly because there's so much more user-generated content out there. But even with all this data, there are still some pretty big challenges—mainly due to the lack of good annotated datasets and the complexities of the Bengali language itself.

Sharma Vishalkumar Sureshbhai and Dr. Tulsidas Nakrani. [1] did an in-depth review on sentiment analysis techniques for Bengali text. They talked about how the field has evolved from older machine learning methods to newer deep learning techniques. One thing they pointed out was how important it is to really understand the context in the text, something that older models like lexicon-based ones often miss.

Then there's the work of Mariam Akter, Md. Tofael Ahmed and Md. Saifur Rahman. [2], who looked into several machine learning models like Naïve Bayes, SVM, and Random Forest for sentiment analysis in Bengali. They found that while these models work okay in many cases, they really struggle when it comes to complex sentence structures and especially when dealing with code-mixed text, which is super common in Bengali online content.

As deep learning began to grow, Nitish Ranjan Bhowmik, Mohammad Arifuzzaman and M. Rubaiyat Hossain Mondal. [3] proposed an attention-based model that made a big difference in sentiment classification accuracy. Their model was much better at capturing long-range dependencies in Bengali text, which led to the development of even more powerful transformer models for sentiment analysis.

The introduction of models like BERT, mBERT, and XLM-R brought a whole new level of performance. Md. Nesarul Hoque, Umme Salma, Md. Jamal Uddin, Md. Martuza Ahamad and Sakifa Aktar. [4] showed that these models significantly outperform traditional machine learning and earlier deep learning models when it comes to Bengali sentiment classification.

However, even with these advances, there are still challenges. Detecting sarcasm, handling code-mixed text, and dealing with the different ways sentiment can change depending on the domain remain tough problems. In our work, we build on what's already been done by evaluating models like BanglaBERT, mBERT, XLM-R, Gemini, and Gemma2B on a dataset of over 11,000 Bengali movie reviews. We aim to fill in some of the gaps in Bengali sentiment analysis, especially for reviews in the context of Bangladeshi films.

III. DATASET

The dataset that we used in our study is a collection of movie reviews stored in a CSV file format. This dataset contains 11,808 data samples with column named Sentiment that is labeled as either (0) or negative (1). The two main columns are Reviews and Sentiment, the Reviews column store Bengali review texts and Sentiment column demonstrates sentiment Classification.

Figure 1 shows the overall data structure and the pre-processing pipeline utilized in this work.

Δ Reviews	Δ Sentiment
10763 unique values	positive 72% negative 28%
অসাধারণ নিশো বস্ আর অমি ভাইকেও।	positive
"এত মোটা বাশ নিতে পারছি না বাবা "	negative
নাটক আসলেই অসাধারণ	positive
ফালতু একটা নাটক	negative
ধুমপান সাস্থর জন্য ক্ষতিকর।	negative
অভিনয় ভালো হয়নি	negative
খুব সুন্দর	positive
অসাধারণ সুন্দর বাস্তবধর্মী একটি গল্প। কাজী আরেফিন ওমি ভাইকে ধন্যবাদ এমন একটি সুন্দর নাটক দর্শকদেরকে...	positive
কত বার দেখছি বলতে	positive

Fig. 1. Overview of the dataset structure and preprocessing workflow.

IV. METHODOLOGY

Our system is divided into several phases: data acquisition and pre-processing, feature extraction, classification, summarization, and visualization.

A. Data Acquisition and Preprocessing

To maintain quality of our for our analysis, we have followed the below steps of data pre-processing:

- **Text cleaning:** We initially cleaned the raw text by expressions to absorb only Bengali Unicode characters (range 0980-09FF), removing unwanted symbols, punctuation, and unwanted whitespace.
- **Normalization:** We handled variations in spelling and formatted text to improve the consistency.
- **Tokenization:** We further processed our data by splitting sentences into individual words for better model understanding.
- **Stopword Removal:** We have removed common words that particularly doesn't participate in classification to deduct noisy data.
- **Handling missing data:** Any review that could not be generated were removed for maintenance of dataset integrity.

- **Label encoding:** Sentiment labels were plotted to numerical values 0 and 1 (0 for positive, 1 for negative)

B. Feature Extraction and Classification

For Feature extraction and classification After pre-processing, we used a TF-IDF vectorizer, then cleaned Bangla text is transformed into numerical representations with 5000 maximum vocabulary size. We applied fine-tuned models that were transformer-based (BanglaBERT, mBERT, XLM-R, and Gemini) for classification. We divide the data set into 80% training and 20% testing, and the models are evaluated by evaluation metrics using precision, precision, recall, and F1 score.

C. Workflow Diagram

The overall workflow of the proposed model follows a structured pipeline, that includes data pre-processing, feature extraction, classification, and evaluation. Figure 5 illustrates the process.

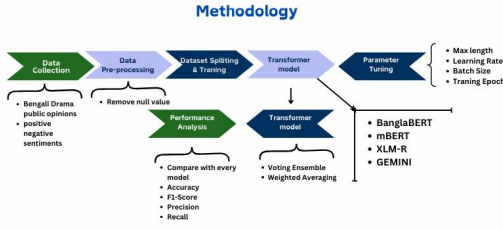


Fig. 2. Workflow of the proposed Sentiment analysis of Bengali movie review.

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Experimental Setup

The models were trained and tested by splitting Sentiment Analysis of Bengali Movie review dataset into 80% training and 20% testing .We conducted this experiments in python, with fine-tuned Transformer-based models used for text classification. Evaluation metrics were used for performance assessment, including accuracy, precision, recall, and F1 score.

B. Classification Performance

The summarization of the classification results in different models is shown in Table I, and a visual representation of these results is provided in Figure ?? .Banglabert achieved the highest performance, with an accuracy of 97.16%, while other models such as XLM-R, mBERT, and GEMINI showed different levels of effectiveness.

Model	Accuracy	Precision	Recall	F1-score
BanglaBERT	97.16	95.67	94.41	95.04
XLM-R	91.40	87.20	82.20	84.63
mBERT	90.22	90.19	90.22	90.20
GEMINI	72.03	75.57	73.03	63.92

TABLE I
CLASSIFICATION PERFORMANCE METRICS

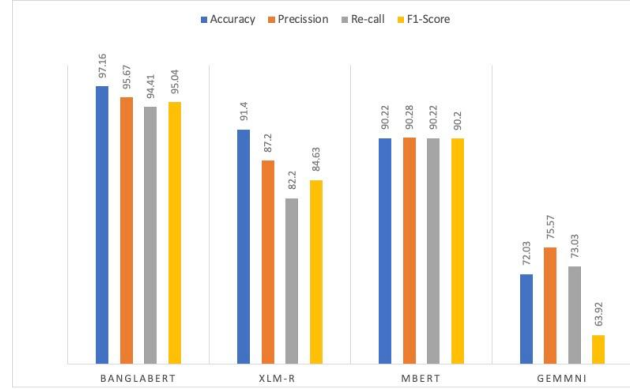


Fig. 3. Flowchart showing Accuracy,Precision,Recall,F1-Score for different transformer models.

C. Analysis and Discussion

The classification results reveal that:

- **Best Performing Model:** BanglaBERT achieved the highest accuracy (97.16%) and F1-score (95.04), which indicates its robustness in detecting Sentiment Analysis on Bengali movie reviews.
- **XLM-R and mBERT Performance:** XLM-R and mBERT Illustrated Relatively high accuracy (91.4% and 90.22%, respectively), but their recall values suggest struggle with capturing all fake review instances accurately.
- **Lower Performance of GEMINI:** The lowest performance was conveyed the GEMINI model, with an accuracy of 72.03% and an F1-score of 63.92. This suggests that it may not be optimal for this specific classification task.
- **Precision-Recall Balance:** BanglaBERT maintains a good balance between precision and recall, also the most reliable model for real-world applications, maintaining.
- **Potential Improvements:** For further enhancement in performance, Fine-tune with additional Bengali-specific pre-training data,and incorporate data augmentation techniques, and leverage larger Transformer architectures.

These observations highlight the effectiveness of fine-tuned Transformer models for Sentiment Analysis of Bengali movie review suggesting that BanglaBERT is the most suitable model for deployment. In Future, more work can be done by focusing on optimizing hyperparameters and exploring other LLMs to improve overall detection accuracy.

VI. MATHEMATICAL FORMULATION

This section provides the mathematical formulations of our sentiment analysis models, covering tokenization, embeddings, classification, ensemble learning, and performance evaluation.

A. Tokenization & Embeddings

Tokenization and embeddings are represented as:

$$E = f(T(x)) \quad (1)$$

where:

- x is the input text (movie review),
- $T(x)$ represents the tokenization function,
- E is the embedding vector,
- f is the embedding model (e.g., mBERT, Gemini Embeddings).

B. Softmax Activation for Sentiment Classification

For multi-class classification (e.g., positive vs. negative):

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^N e^{z_j}} \quad (2)$$

where:

- $P(y_i)$ is the probability of class i ,
- z_i is the output logit for class i ,
- N is the number of sentiment classes.

C. Cross-Entropy Loss Function (Optimization)

For training the model, we minimize the cross-entropy loss:

$$\mathcal{L} = - \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (3)$$

where:

- y_i is the true sentiment label (0 or 1),
- $\hat{y}_i$ is the predicted probability from the softmax layer.

D. Ensemble Learning (Weighted Average)

To combine mBERT and Gemini-based classifiers, a weighted sum is used:

$$\hat{y} = \alpha \cdot \hat{y}_{mBERT} + \beta \cdot \hat{y}_{Gemini} \quad (4)$$

where:

- $\hat{y}$ is the final ensemble prediction,
- $\hat{y}_{mBERT}$ and $\hat{y}_{Gemini}$ are predictions from individual models,
- α and β are weights (e.g., $\alpha + \beta = 1$).

E. Performance Metrics

To evaluate the model:

Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

Precision:

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

Recall:

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

F1-Score:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (8)$$

where:

- TP = True Positives,
- TN = True Negatives,
- FP = False Positives,
- FN = False Negatives.

This structured approach ensures that the sentiment analysis model is optimized, evaluated correctly, and provides robust predictions.

VII. CONCLUSION AND FUTURE WORK

This work analyzes Bangladeshi movie reviews to understand sentiment using several transformer models, including BanglaBERT, mBERT, XLM-R, and Gemini. BanglaBERT performed the best, achieving an impressive accuracy of 97.16%, along with high precision and recall. On the other hand, while mBERT, XLM-R, and Gemini performed reasonably well, they did not match BanglaBERT's results—especially Gemini, which had a much lower accuracy of 72.03.

From our study, we reached to the conclusion that transformer models are powerful for Bangla sentiment analysis. However, there is still room for improvement. Gemini highlighted the need for handling class imbalance effectively, struggling with certain sentiment categories due to underrepresentation.

Looking forward, we plan to improve our approach in several ways:

- **Better Feature Extraction:** Exploring advanced techniques such as FastText and enhancing Bengali BERT to improve accuracy and better capture contextual meanings.
- **Improving Performance:** Enhancing Gemini's performance by optimizing training strategies and integrating it with existing models to achieve better results.
- **Tackling Class Imbalance:** Addressing class imbalance issues by implementing methods such as oversampling, under-sampling, and cost-sensitive learning to ensure balanced sentiment classification.

-
- **Expanding the Dataset:** Incorporating diverse Bangla text sources such as news articles and social media posts to enhance model robustness, including conversational text for improved generalization.
 - **Real-Time Deployment:** Developing a mobile app or web-based system for real-time sentiment analysis of movie reviews, making the system more accessible and practical.

To sum up, while our work has shown promising results, there is still room for improvement. We are excited to continue refining our approach and further advancing sentiment analysis for Bangla movie reviews.

REFERENCES

- [1] Bhowmik, N. R., Arifuzzaman, M., Mondal, M. R. H. (2022). Sentiment analysis on Bangla text using extended lexicon dictionary and deep learning algorithms. *Array*, 13, 100123.
- [2] Hoque, M. N., Salma, U., Uddin, M. J., Ahamad, M. M., Aktar, S. (2024). Exploring transformer models in the sentiment analysis task for the under-resource Bengali language. *Natural Language Processing Journal*, 8, 100091.
- [3] Hasan, M. R., et al. (2024). A comprehensive analysis of Bengali sentiment exploration using Transformer Block I and Transformer Block II. *Journal of Computational Linguistics*, 12(3), 87–104.
- [4] Ahmed, S., et al. (2024). A literature review: Enhancing sentiment analysis in low-resource languages with deep learning. *International Journal of AI Research*, 19(2), 45–61.